Capstone Project One

Library Management System

Overview

Create a simple Library Management System using Python and Object-Oriented Programming (OOP). The program should store data in CSV files so that information is not lost when the program closes.

Learning Objectives

- Create and use classes in Python
- Use inheritance (Student and Faculty)
- Use lists and dictionaries
- Read and write CSV files
- Handle invalid input using try/except

Required Classes

Class	Purpose
Book	Store ISBN, title, author, total copies, available copies
Member	Store member ID, name, borrowed books
Student	Member type with borrowing limit 3
Faculty	Member type with borrowing limit 5
Library	Manage books and members

Functions to Implement

1. Add Book
2. Register Member (Student or Faculty)
3. Issue Book
4. Return Book
5. Search Book by title or author
6. View all books
7. Save data in CSV files

CSV Files

Store data in.

- books.csv
- members.csv
- borrowed.csv

Example Data (Sample CSV Content)

You can start with the following simple sample data. Your program should be able to load, update, and save this data.

books.csv

isbn,title,author,total_copies,available_copies
9780134685991,Effective Python,Brett Slatkin,5,5
9781492056355,Fluent Python,Luciano Ramalho,3,2
9781593279288,Python Crash Course,Eric Matthes,4,4
9781449355739,Learning Python,Mark Lutz,2,1

members.csv

member_id,name,member_type
S001,Alice Johnson,Student
S002,Bob Smith,Student
F001,Dr. Carol Lee,Faculty
F002,Prof. David Kim,Faculty

borrowed.csv

member_id,isbn
S001,9781492056355
F001,9781449355739

Note: In the sample above, Alice (S001) has borrowed "Fluent Python" and Dr. Carol (F001) has borrowed "Learning Python". Available copies are already reduced accordingly.

Error Handling

Use try/except to prevent the program from crashing. User defined Exception.

- Book not found
- No copies available
- Member not found
- Borrowing limit reached

Deliverables

- All Python files
- CSV data files
- Documentation

Mark Division

Criteria	Marks
Use of classes and inheritance	15
Functions working correctly	35
CSV file saving/loading	20
Code readability	10
Documentation	20

Student Test Cases (Must Pass)

Students should run the following test cases and make sure the program produces the expected result.

TC	Feature	Input / Action	Expected Result
TC-01	Add Book	Add a new book with ISBN 1111 and 2 copies	Book is added and saved in books.csv
TC-02	Register Student	Register student S010, Ram	Member appears in members.csv
TC-03	Register Faculty	Register faculty F010, Hari	Member appears in members.csv
TC-04	Issue Available Book	Issue Effective Python to S010	available_copies decreases by 1 and entry added to borrowed.csv
TC-05	Return Book	Return Effective Python from S010	available_copies increases by 1 and entry removed from borrowed.csv
TC-06	Search by Title	Search 'Python Crash Course'	Correct book information is displayed
TC-07	Search by Author	Search 'Mark Lutz'	Learning Python is displayed
TC-08	Book Not Found	Search 'Unknown Book'	Program shows 'Book not found' without crashing
TC-09	Member Not Found	Issue a book to S999	Program shows 'Member not found'
TC-10	No Copies Available	Set available copies to 0 and issue the book	Program shows 'No copies available'
TC-11	Student Limit	Issue 4 books to a student	Fourth issue is rejected with 'Borrowing limit reached'
TC-12	Faculty Limit	Issue 6 books to a faculty member	Sixth issue is rejected with 'Borrowing limit reached'
TC-13	Data Persistence	Close and run the program again	Books, members, and borrowed records are loaded correctly

Submission Checklist

- Program runs without crashing.
- All 13 test cases pass.

- CSV files are created inside the data/ folder.
- Code uses classes and inheritance.
- Proper documentation of your work along with screenshot of your passed test cases within a documentation.